

PRACTICAL NO. 1: FREQUENCY AND CHARTS

- a. Recode the variable 'value of the house' as 1-100-150, 2-150-200 and so on. Compute the frequency table for age group, education, religion and number of years in the company.

Solution:

Frequency Table

Age group					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	20-29 year	20	13.3	13.3	13.3
	30-39 year	41	27.3	27.3	40.7
	40-49 year	38	25.3	25.3	66.0
	above 49 year	50	33.3	33.3	99.3
	5	1	.7	.7	100.0
	Total	150	100.0	100.0	

Education of the employee					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	High school	6	4.0	4.0	4.0
	same college	16	10.7	10.7	14.7
	College graduate	42	28.0	28.0	42.7
	Master's degree	61	40.7	40.7	83.3
	Doctoral degree	25	16.7	16.7	100.0
	Total	150	100.0	100.0	

Religious denomination					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Catholic	28	18.7	18.7	18.7
	Protestant	40	26.7	26.7	45.3
	Jewish	41	27.3	27.3	72.7
	Other	41	27.3	27.3	100.0
	Total	150	100.0	100.0	

Number of year with the company					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Less than 5 year	17	11.3	11.3	11.3
	5 and upto 10 year	34	22.7	22.7	34.0
	10 and upto 15 year	47	31.3	31.3	65.3
	15and upto 20 year	33	22.0	22.0	87.3
	over 20 year	19	12.7	12.7	100.0
	Total	150	100.0	100.0	

Interpretation:

Age Group: Most of the employees are older, with the largest group being "above 49 years old" (33.3%). Younger employees are rare (only 13.3% are under 30).

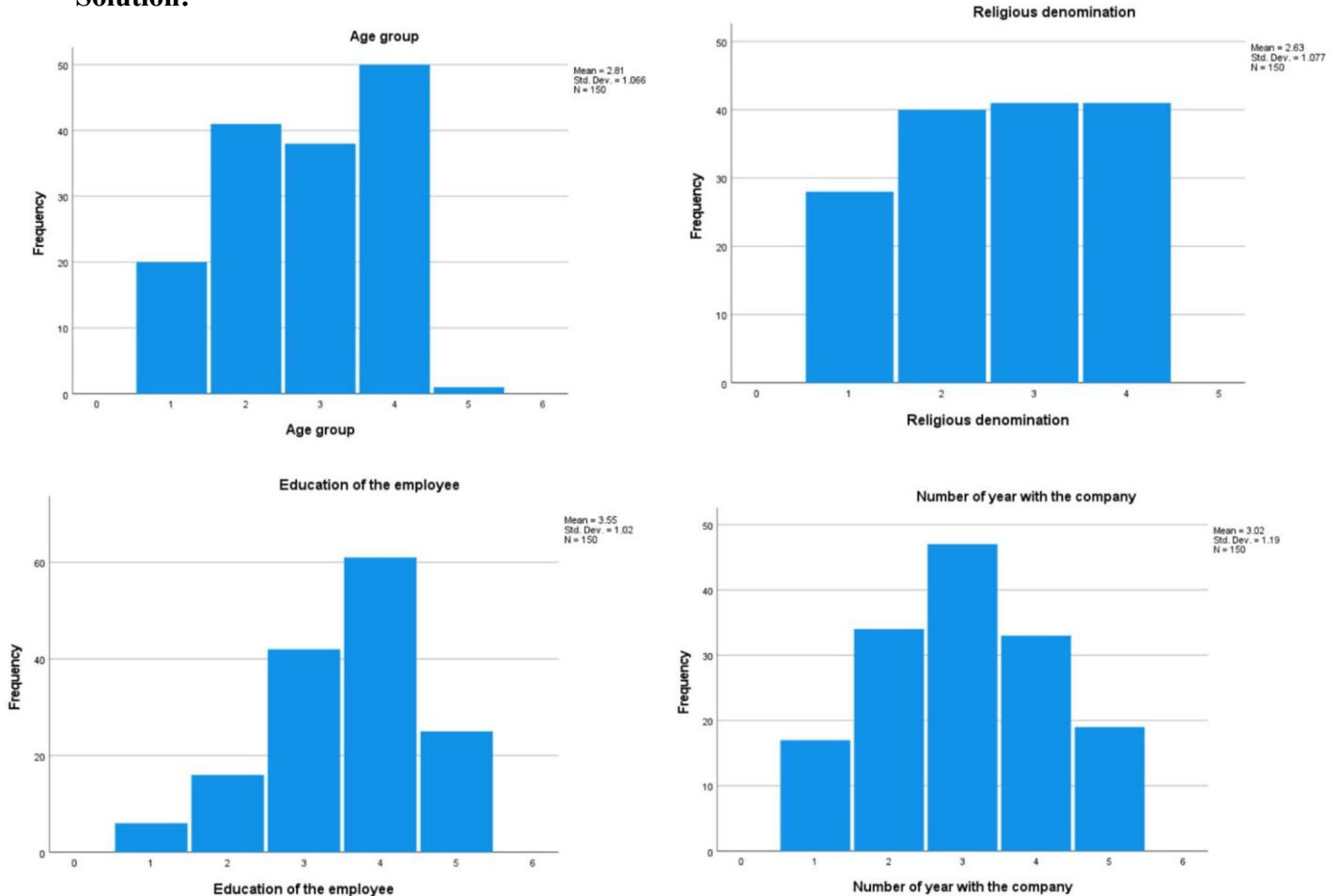
Education: The employees are highly educated. Most people have a Master's degree (40.7%) while few people stopped after high school (only 4%).

Religion: The company has a diverse religious mix. Protestants, Jewish employees, and those from "Other" religions each make up about 27%. Catholics are slightly fewer, at 18.7%.

Years in the Company: Tenure leans toward medium-to-long service. The largest group (31.3%) has been with the company for 10–15 years. There are more long-term employees (over 20 years) than newcomers (under 5 years).

b. Construct the histogram for age group, education and religion and number of years in the company.

Solution:



Interpretation:

Age Group (Mean = 2.81, SD = 1.07): The average age group falls between 30-39 and 40-49 years. The standard deviation shows moderate spread, meaning employees are distributed across all age ranges.

Religious Denomination (Mean = 2.63, SD = 1.08): The mean has no real meaning since religion is categorical. The histogram simply shows the four groups have nearly equal frequencies.

Education (Mean = 3.55, SD = 1.02): The average education level is between College Graduate and Master's degree. The low standard deviation indicates most employees cluster around this high education level.

Number of Years in Company (Mean = 3.02, SD = 1.19): The average tenure falls near the 10-15 year category. The slightly higher standard deviation shows more variability in tenure compared to other variables.

- c. Show stem and leaf display for salary, value of the house, Distance in miles from house to work, Sick days taken during the previous year.

Solution:

Salary in thousand of dollar Stem-and-Leaf Plot

Frequency	Stem &	Leaf
1.00	2 .	7
8.00	3 .	22257778
16.00	4 .	0002255555777899
16.00	5 .	0002222245556777
20.00	6 .	000000022235555556888
27.00	7 .	000000011222223555555557779
29.00	8 .	000000000000001222223555555557
17.00	9 .	00000001222455557
7.00	10 .	0000067
5.00	11 .	00002
4.00	12 .	0005

```
Stem width:      10
Each leaf:       1 case(s)
```

Sick days taken during the previous year Stem-and-Leaf Plot

[illegible]

```
Stem width:      10
Each leaf:       1 case(s)
```

Distance in miles from house to work Stem-and-Leaf Plot

[illegible]

Stem width: 10
Each leaf: 1 case(s)

Value of the house in thousand of dollar Stem-and-Leaf Plot

Frequency	Stem &	Leaf
.00	1 .	
3.00	1 .	233
3.00	1 .	445
9.00	1 .	666666777
13.00	1 .	8888889999999
17.00	2 .	0000000000111111
7.00	2 .	2223333
19.00	2 .	44444444455555555
15.00	2 .	6666666666666777
21.00	2 .	888888888888888999999
16.00	3 .	0000000000000111
12.00	3 .	222222222233
4.00	3 .	4455
6.00	3 .	666667
4.00	3 .	8889
1.00	4 .	0

```
Stem width:      100
Each leaf:       1 case(s)
```

Interpretation:

Salary: Positively skewed (tail extends toward higher values)

Sick Days: Positively skewed (concentration on lower end, tail toward upper end)

Distance: Approximately symmetric (values evenly distributed around center)

House Value: Approximately symmetric (balanced distribution around midpoint)

PRACTICAL NO. 2: DESCRIPTIVE STATISTICS

- a. Draw descriptive statistics of all employee, Male employee, female employee for the salary, value of the house, Distance in miles from house to work, Sick days taken during the previous year.

Solution:

Sex of employee		Statistic	Std. Error
Salary in thousand of dollar	Female	Mean	60.28
		95% Confidence Interval for Mean	2.213
		Lower Bound	55.85
		Upper Bound	64.71
		5% Trimmed Mean	60.17
		Median	60.00
		Variance	293.868
		Std. Deviation	17.143
		Minimum	27
		Maximum	97
		Range	70
		Interquartile Range	28
		Skewness	.133
		Kurtosis	.309
	Male	Mean	80.06
		95% Confidence Interval for Mean	2.079
		Lower Bound	75.92
		Upper Bound	84.19
		5% Trimmed Mean	80.09
		Median	80.00
		Variance	389.019
		Std. Deviation	19.724
		Minimum	32
		Maximum	125
		Range	93
		Interquartile Range	22
		Skewness	-.058
		Kurtosis	.254

Value of the house in thousand of dollar	Female	Mean	235.58	7.207
		95% Confidence Interval for Mean	221.16	
		Lower Bound	250.00	
		Upper Bound	235.89	
		5% Trimmed Mean	240.00	
		Median	3116.620	
		Variance	55.827	
		Std. Deviation	120	
		Minimum	360	
		Maximum	240	
		Range	86	
		Interquartile Range	-.075	.309
		Skewness	-.629	.608
		Kurtosis	270.77	6.528
	Male	Mean	257.80	
		95% Confidence Interval for Mean	283.74	
		Lower Bound	270.59	
		Upper Bound	280.00	
		5% Trimmed Mean	3835.057	
		Median	61.928	
		Variance	150	
		Std. Deviation	400	
		Minimum	250	
		Maximum	90	
		Range	-.029	.254
		Interquartile Range	-.717	.503
		Skewness		
		Kurtosis		

Sick days taken during the previous year	Female	Mean	8.37	.675
		95% Confidence Interval for Mean	7.02	
		Lower Bound	9.72	
		Upper Bound	8.30	
		5% Trimmed Mean	8.50	
		Median	27.321	
		Variance	5.227	
		Std. Deviation	0	
		Minimum	20	
		Maximum	20	
		Range	7	
		Interquartile Range	-.168	.309
		Skewness	-.627	.608
		Kurtosis	8.21	.430
	Male	Mean	7.36	
		95% Confidence Interval for Mean	9.07	
		Lower Bound	8.19	
		Upper Bound	8.00	
		5% Trimmed Mean	16.640	
		Median	4.079	
		Variance	0	
		Std. Deviation	20	
		Minimum	20	
		Maximum	5	
		Range	-.084	.254
		Interquartile Range	.367	.503
		Skewness		
		Kurtosis		

Distance in miles from house to work	Female	Mean	17.65	1.063
		95% Confidence Interval for Mean	15.52	
		Lower Bound	19.78	
		Upper Bound	17.48	
		5% Trimmed Mean	16.50	
		Median	67.757	
		Variance	8.231	
		Std. Deviation	4	
		Minimum	35	
		Maximum	31	
		Range	13	
		Interquartile Range	.265	.309
		Skewness	-.797	.608
		Kurtosis	15.78	.678
	Male	Mean	14.43	
		95% Confidence Interval for Mean	17.13	
		Lower Bound	15.52	
		Upper Bound	15.00	
		5% Trimmed Mean	41.388	
		Median	6.433	
		Variance	2	
		Std. Deviation	32	
		Minimum	30	
		Maximum	10	
		Range	.613	.254
		Interquartile Range	-.004	.503
		Skewness		
		Kurtosis		

Interpretation:

Males have higher average salary (80.06k vs 60.28k) and house value (270.77k vs 235.58k) than females. Females take slightly more sick days (8.37 vs 8.21) and travel farther to work (17.65 vs 15.78 miles). Males show greater variability in salary and house value, while females show more variability in sick days and distance. All distributions are approximately symmetric.

- b. The coefficient of variation of the salaries for male and female and find whose salary is most uniformly distributed.

Solution:**Formula:**

$$CV = \left(\frac{\text{Std. Deviation}}{\text{Mean}} \right) \times 100$$

For Female:

- Mean = 60.28
- Std. Deviation = 17.143

$$CV_{\text{Female}} = \left(\frac{17.143}{60.28} \right) \times 100 = 28.44\%$$

For Male:

- Mean = 80.06
- Std. Deviation = 19.724

$$CV_{\text{Male}} = \left(\frac{19.724}{80.06} \right) \times 100 = 24.64\%$$

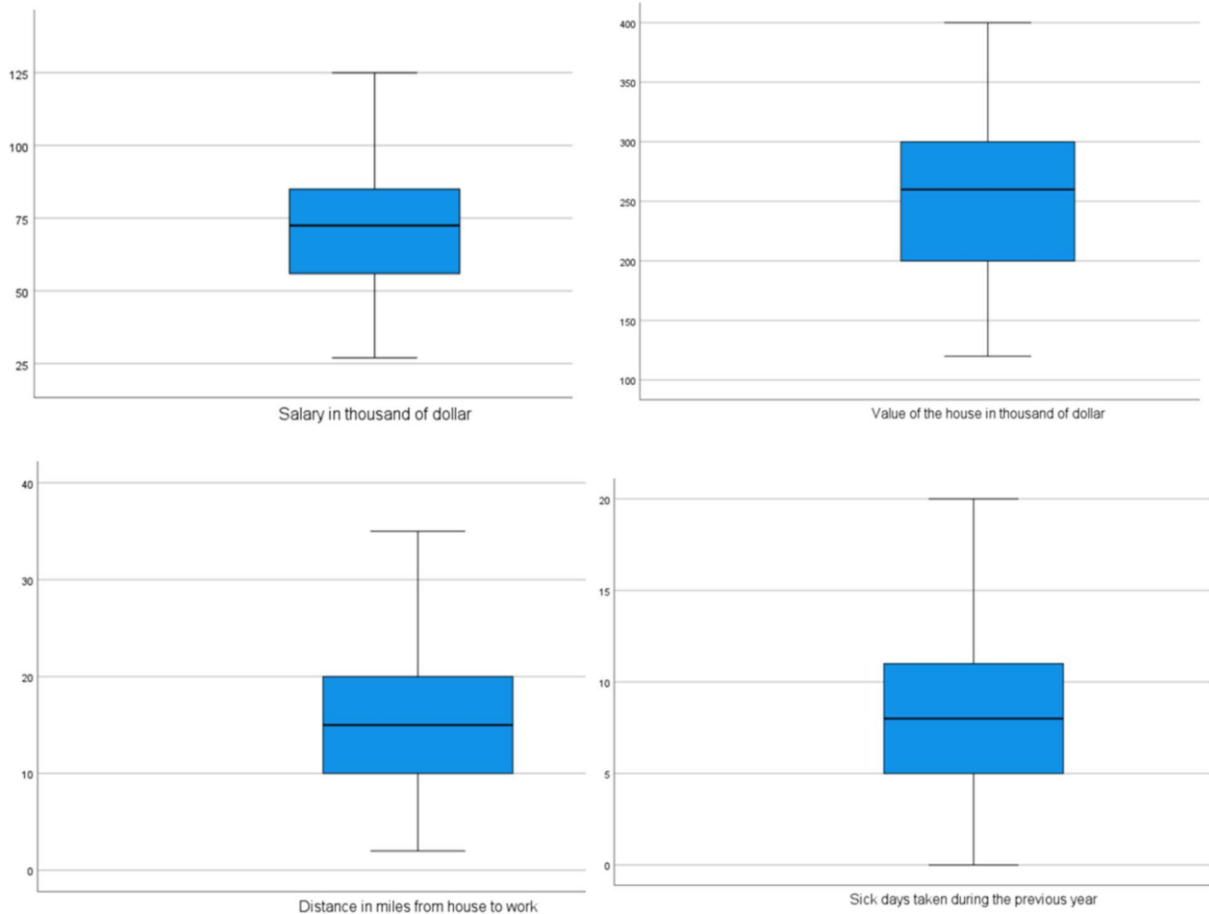
Conclusion:

- Male CV = 24.64%
- Female CV = 28.44%

Since lower CV indicates more uniformity (less relative variability), male salaries are more uniformly distributed than female salaries.

- c. Box plot for salary, value of the house, distance in miles from house to work, Sick days taken during the previous year.

Solution:



Interpretation:

- **Salary:** The distribution appears slightly positively skewed (right-skewed) since the upper whisker is longer and values extend further above the median.
- **House Value:** The distribution is positively skewed, indicating a few houses with much higher values than the majority.
- **Distance to Work:** The distribution is positively skewed, showing that while most people commute moderate distances, a few travel much longer distances.
- **Sick Days:** The distribution is slightly positively skewed, meaning most employees take relatively fewer sick days, with a few taking significantly more.

PRACTICAL NO. 3: TEST OF HYPOTHESIS

a. One sample test:

- i. Test the hypothesis that the average salary of the whole employees of the company is 75 thousand dollars and 50 thousand dollar and interpret.

Solution:

- $H_0: \mu = 75$
- v/s $H_1: \mu \neq 75$
- Level of significance: $\alpha = 0.05$

One-Sample Statistics				
	N	Mean	Std. Deviation	Std. Error Mean
Salary in thousand of dollar	150	72.15	21.052	1.719

One-Sample Test						
Test Value = 75						
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Salary in thousand of dollar	-1.660	149	.099	-2.853	-6.25	.54

Interpretation: The t value is -1.680. Since p value = 0.999 > 0.05. Hence we do not reject H_0 . There is strong evidence to say that the average salary does not differ from 75 thousand dollars.

- $H_0: \mu = 50$
- v/s $H_1: \mu \neq 50$

One-Sample Statistics				
	N	Mean	Std. Deviation	Std. Error Mean
Salary in thousand of dollar	150	72.15	21.052	1.719

One-Sample Test						
Test Value = 50						
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Salary in thousand of dollar	12.884	149	<.001	22.147	18.75	25.54

Interpretation: The t value is 12.884. Since p value < 0.001 < 0.05. Hence we reject H_0 . There is strong evidence to say that the average salary differs from 50 thousand dollars.

- ii. Test the hypothesis of 10% of the data set 1 that the average salary is 75 thousand dollars and 50 thousand dollar and interpret.

Solution:

- $H_0: \mu = 75$
- v/s $H_1: \mu \neq 75$

One-Sample Statistics				
	N	Mean	Std. Deviation	Std. Error Mean
Salary in thousand of dollar	17	80.88	16.763	4.066

One-Sample Test						
Test Value = 75						
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Salary in thousand of dollar	1.447	16	.167	5.882	-2.74	14.50

Interpretation: The t value is 1.447. Since p value = 0.167 > 0.05. Hence we do not reject H_0 . There is strong evidence to say that the average salary does not differ from 75 thousand dollars.

- $H_0: \mu = 50$
- v/s $H_1: \mu \neq 50$

One-Sample Statistics				
	N	Mean	Std. Deviation	Std. Error Mean
Salary in thousand of dollar	17	80.88	16.763	4.066

One-Sample Test						
Test Value = 50						
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Salary in thousand of dollar	7.596	16	<.001	30.882	22.26	39.50

Interpretation: The t value is 7.596. Since p value < 0.001 < 0.05. Hence we reject H_0 . There is strong evidence to say that the average salary differs from 50 thousand dollars.

- iii. Test the hypothesis of 10% of the data set 1 that the average value of house is 75 thousand dollars and 50 thousand dollar and interpret.

Solution:

- $H_0: \mu = 75$ v/s $H_1: \mu \neq 75$

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
Value of the house in thousand of dollar	17	265.00	67.407	16.349

One-Sample Test

Test Value = 75						
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Value of the house in thousand of dollar	11.622	16	<.001	190.000	155.34	224.66

Interpretation: The t value is 11.622. Since p value $< 0.001 < 0.05$. Hence, we reject H_0 . There is strong evidence to say that the average house value is not equal to 75 thousand dollars.

- $H_0: \mu = 50$
- v/s $H_1: \mu \neq 50$

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
Value of the house in thousand of dollar	17	265.00	67.407	16.349

One-Sample Test

Test Value = 50						
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Value of the house in thousand of dollar	13.151	16	<.001	215.000	180.34	249.66

Interpretation: The t value is 13.151. Since p value $< 0.001 < 0.05$. Hence, we reject H_0 . There is strong evidence to say that the average salary differs from 50 thousand dollars.

- iv. At 0.05 level of significance, is their evidence that average number of sick days taken during the previous year were 10 days. (use id 1 to 25 of the data set1)

Solution:

- $H_0: \mu = 10$
- v/s $H_1: \mu \neq 10$

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
Sick days taken during the previous year	25	8.92	4.873	.975

One-Sample Test

Test Value = 10						
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Sick days taken during the previous year	-1.108	24	.279	-1.080	-3.09	.93

Interpretation: The t value is -1.108. Since p value = 0.279 > 0.05. Hence, we do not reject H_0 . There is strong evidence to say that the average number of sick days is equal to 10 days.

b. Two sample tests:

- i. Test the hypothesis that is there any significance difference between mean salary of male and female employees.

Solution:

- $H_0: \mu_1 = \mu_2$
- v/s $H_1: \mu_1 \neq \mu_2$

Group Statistics					
	Sex of employee	N	Mean	Std. Deviation	Std. Error Mean
Salary in thousand of dollar	Female	60	60.28	17.143	2.213
	Male	90	80.06	19.724	2.079

Independent Samples Test										
		Levene's Test for Equality of Variances			t-test for Equality of Means					
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
Salary in thousand of dollar	Equal variances assumed	.205	.652	-6.331	148	<.001	-19.772	3.123	-25.943	-13.601
	Equal variances not assumed			-6.512	137.894	<.001	-19.772	3.036	-25.776	-13.768

Interpretation: Here $t = -6.331$, $df = 148$. Since $p\text{-value} < 0.001 < 0.05$. Hence, we reject H_0 . There is strong evidence to say that there is significance difference between mean salary of male and female employees.

- ii. Test the hypothesis that is there any significance difference between mean salary of male and female (use id 25 to 50 of the data set1)

Solution:

- $H_0: \mu_1 = \mu_2$
- v/s $H_1: \mu_1 \neq \mu_2$

Group Statistics					
	Sex of employee	N	Mean	Std. Deviation	Std. Error Mean
Salary in thousand of dollar	Female	11	60.45	19.367	5.839
	Male	15	86.60	14.024	3.621

Independent Samples Test										
		Levene's Test for Equality of Variances			t-test for Equality of Means					
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
Salary in thousand of dollar	Equal variances assumed	.935	.343	-4.001	24	<.001	-26.145	6.535	-39.633	-12.658
	Equal variances not assumed			-3.805	17.338	.001	-26.145	6.871	-40.620	-11.671

Interpretation: Here $t = 4.001$, $df = 24$. Since $p\text{-value} < 0.001 < 0.05$. Hence, we reject H_0 . There is strong evidence to say that there is significance difference between mean salary of male and female employees.

- iii. At 0.05 level of significance, is their evidence that average number of sick days taken during the previous year were significance difference between male and female. (10% of the data set 1).

Solution:

- $H_0: \mu_1 = \mu_2$
- v/s $H_1: \mu_1 \neq \mu_2$

Group Statistics					
	Sex of employee	N	Mean	Std. Deviation	Std. Error Mean
Sick days taken during the previous year	Female	6	7.50	5.683	2.320
	Male	15	7.27	4.148	1.071

Independent Samples Test										
		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
Sick days taken during the previous year	Equal variances assumed	.934	.346	.105	19	.918	.233	2.223	-4.420	4.886
	Equal variances not assumed			.091	7.241	.930	.233	2.556	-5.769	6.236

Interpretation: Here $t = 0.105$, $df = 19$. Since $p\text{-value} = 0.918 > 0.05$. Hence, we do not reject H_0 . There is strong evidence to say that there is no significance difference between average number of sick days taken during the previous year by male and female.

c. One way ANOVA test:

- i. Test the hypothesis that is there any significance difference between mean salary of employees with education level.

Solution: $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$

v/s H_1 : at least one is different

Descriptives

Salary in thousand of dollar

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
High school	6	36.83	9.745	3.978	26.61	47.06	27	55
same college	16	64.69	16.796	4.199	55.74	73.64	42	92
College graduate	42	69.40	20.035	3.092	63.16	75.65	32	110
Master's degree	61	75.25	20.422	2.615	70.02	80.48	35	125
Doctoral degree	25	82.44	18.171	3.634	74.94	89.94	49	120
Total	150	72.15	21.052	1.719	68.75	75.54	27	125

ANOVA

Salary in thousand of dollar

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	11922.912	4	2980.728	7.987	<.001
Within Groups	54111.861	145	373.185		
Total	66034.773	149			

Interpretation: Here, $F = 7.987$. Since $p\text{-value} < 0.001 < 0.05$. Hence, we reject H_0 . There is significant evidence that the mean salary differs across education levels.

- ii. Test the hypothesis that is there any significance difference between value of house with number of bed room in the house.

Solution:

Descriptives

Value of the house in thousand of dollar

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
1	3	156.67	35.119	20.276	69.43	243.91	120	190
2	21	173.05	25.384	5.539	161.49	184.60	130	230
3	74	254.19	45.735	5.317	243.59	264.79	162	360
4	51	298.24	49.212	6.891	284.39	312.08	180	400
5	1	380.00	.	.	.	.	380	380
Total	150	256.69	61.838	5.049	246.72	266.67	120	400

ANOVA

Value of the house in thousand of dollar

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	280625.746	4	70156.437	35.183	<.001
Within Groups	289138.147	145	1994.056		
Total	569763.893	149			

Interpretation: Here, $F = 35.183$. Since $p\text{-value} < 0.001 < 0.05$. Hence, we do not reject H_0 . There is significant evidence that the mean house value differs across number of bedrooms.